



TDM INSIGHT

TDM Insight

Vancomycin Therapeutic Drug Monitoring — Mobile Application

Mobile App Development • Bachelor of Data Science

Albukhary International University (AIU) — School of Computing & Informatics

PROJECT TEAM

1

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UI/UX & Frontend Developer

2

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Application Integration, Testing & Documentation

Agenda

What we'll cover in this walkthrough of TDM Insight.

1

Introduction

The clinical problem TDM Insight addresses

2

Objectives & Solution

Goals and the approach we implemented

3

Key Features

What the application does for the user

4

TDM Workflows

Pre-dose, post-dose and pre+post methods

5

PK Calculations

The pharmacokinetic engine behind results

6

Architecture & Tech Stack

How the app is built

7

App Walkthrough

Screens from patient entry to results

8

Status & Next Steps

Where the project stands, what's next

Why Vancomycin Monitoring Is Hard

Vancomycin is an antibiotic that requires careful therapeutic drug monitoring (TDM) — the right exposure is essential for treatment to work while keeping toxicity risk low.

Manual TDM calculations require pulling together many inputs at once — patient characteristics, renal function, dosing history, infusion duration, and sampled drug concentrations — making the process slow and error-prone.



Our Goal

Provide a structured mobile app that simplifies the educational demonstration of Vancomycin TDM calculations through a guided, user-friendly interface.

Inputs Needed for a Manual Calculation

- Patient characteristics (age, sex, weight)
- Serum creatinine (renal function)
- Dosage & dosing interval
- Infusion duration
- Drug concentrations (pre/post)
- Sampling times

Objectives & Implemented Solution

TDM Insight focuses on eight core outcomes, delivered through a guided, three-step assessment flow.



What the Application Does

1

Multiple TDM Workflows

Pre-dose, post-dose, and pre+post two-point estimation.

2

Patient Information

Case name, sex, age, body weight, serum creatinine.

3

Medication Information

Dose, interval, infusion duration, concentrations, sample time.

4

Input Validation

Checks required fields, ranges, and calculation conditions.

5

PK Calculations

CrCl, Ke, half-life, Vd, clearance, peak concentration, AUC₂₄.

6

Calculation Explanation

Expandable, step-by-step view of intermediate values.

7

Offline Application

No internet, backend, cloud DB, or account required.

8

Academic Safety Disclaimer

Clear notice that results are educational, not clinical.

Three Guided TDM Workflows

Pre-Dose

Uses a pre-dose / trough concentration taken before the next dose.

SELECT METHOD → ENTER DATA → VIEW RESULT

Post-Dose

Uses a post-dose concentration and its sampling time after infusion.

SELECT METHOD → ENTER DATA → VIEW RESULT

Pre + Post

Combines both concentrations for a two-point PK estimation.

SELECT METHOD → ENTER DATA → VIEW RESULT

Pharmacokinetic Calculation Engine

The TdmEngine estimates the following parameters from validated patient and dosing inputs:

CrCl

Creatinine Clearance

Cockcroft–Gault estimate of renal function

Ke

Elimination Rate Constant

Rate at which the drug is cleared

$t_{1/2}$

Elimination Half-life

Time for concentration to halve

Vd

Volume of Distribution

How the drug distributes in the body

CL

Clearance

Volume of plasma cleared per hour

Cmax

Estimated Peak Concentration

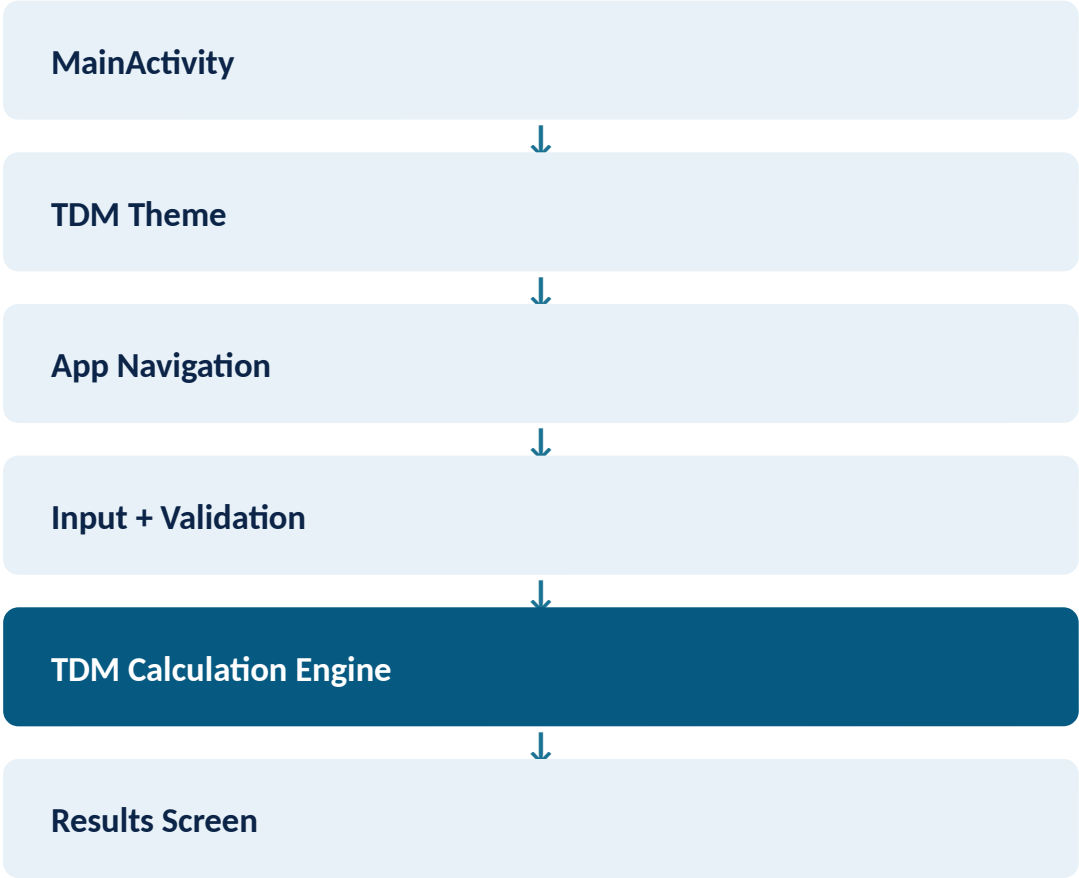
Predicted maximum drug level

AUC24

Area Under the Curve (24h)

Primary result — overall drug exposure

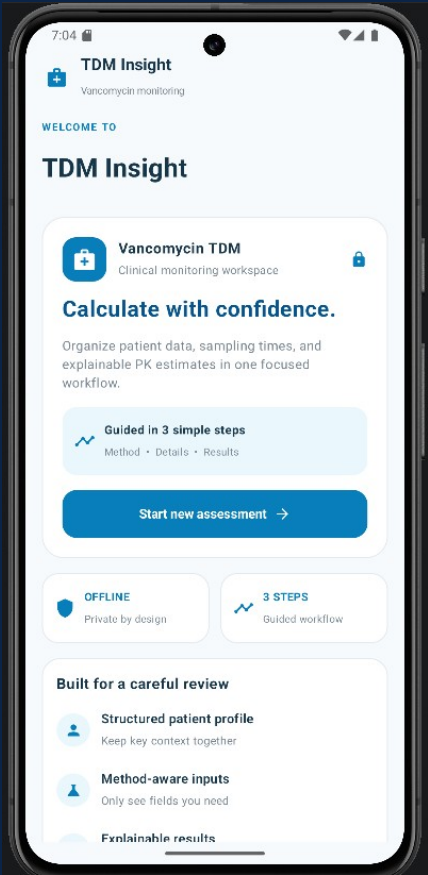
Application Architecture & Tech Stack



TECHNOLOGY STACK

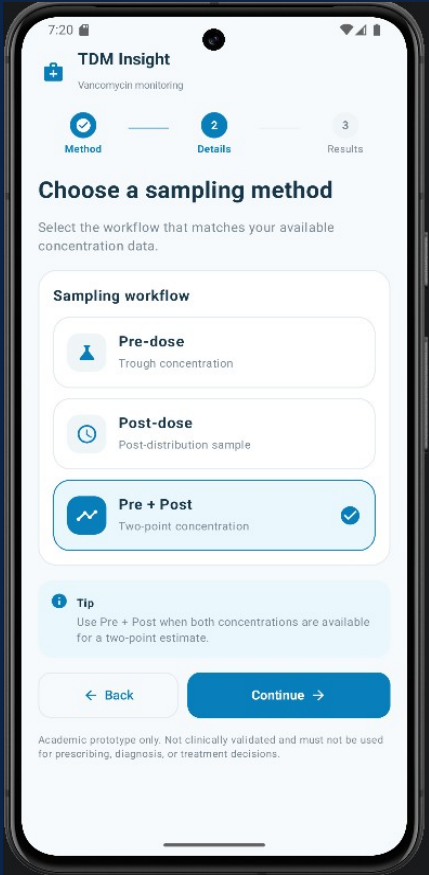
	Kotlin	Main programming language
	Jetpack Compose	User interface development
	Material 3	UI components and design
	Android Studio / SDK	Development environment
	Gradle Kotlin DSL	Build & dependency management
	Git & GitHub	Version control & collaboration

From Patient Entry to Explainable Results



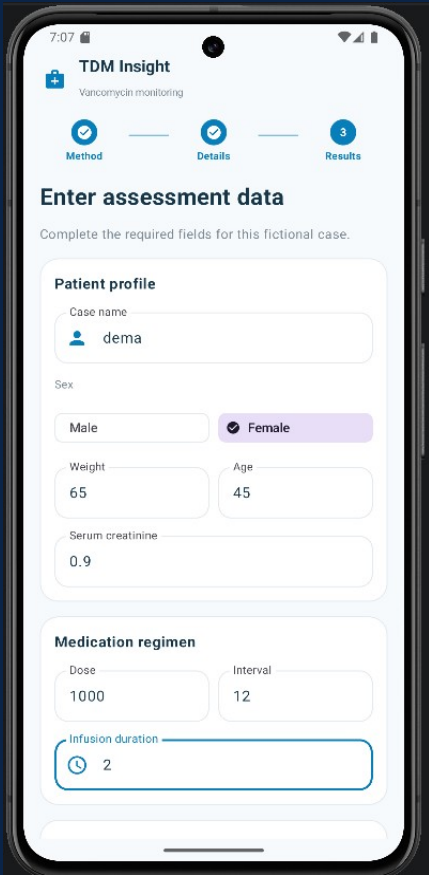
1. Home

Start a new guided assessment



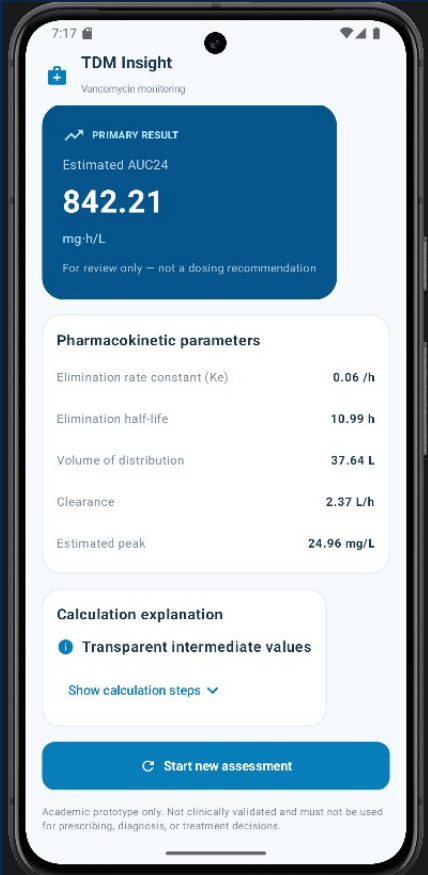
2. Method

Choose Pre / Post / Pre+Post



3. Patient Data

Enter patient & dosing details



4. Results

AUC24 & PK parameters, explained

Project Status & Future Improvements

CURRENT STATUS

Completed Academic Prototype · v1.0

- ✓ Native Android app (Jetpack Compose)
- ✓ Patient & medication data entry
- ✓ PK calculation engine + AUC24
- ✓ Offline operation
- ✓ Three TDM workflows
- ✓ Input validation
- ✓ Calculation explanation
- ✓ Academic safety disclaimer

FUTURE IMPROVEMENTS

- 1 Local SQLite / Room database
- 2 Assessment history
- 3 Patient case management
- 4 PDF report generation
- 5 Exporting calculation results
- 6 Cloud synchronization
- 7 User authentication
- 8 Additional TDM drug workflows

References & Disclaimer

REFERENCES

- Android Developers Documentation
- Kotlin Documentation
- Jetpack Compose Documentation
- Material Design 3 Documentation
- Cockcroft–Gault equation for creatinine clearance
- Vancomycin TDM guidelines & academic references



Academic Disclaimer

TDM Insight is an academic software prototype developed for educational purposes. The results generated are estimates based on the mathematical models implemented in the application.

They are not clinically validated and must not be used independently for prescribing, diagnosis, treatment, or patient-care decisions. All patient examples used with this application should be fictional or appropriately anonymized.



Thank You

Questions & Discussion Welcome

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